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MILITARY SPECIFICATION

FUZE, POINT DETONATING, M505A1 LOADING, ASSEMBLING, AND PACKING

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for the M505A1 Point Detonating Fuze, for use in 20mm ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–G–2550 — General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL–STD–350 — Jolt Test for Use in Production of Fuzes.

MIL–STD–351 — Jumble Test for Use in Production of Fuzes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

D7258818 — Fuze, PD, M505A1, Assembly.

D7553814 — Cartridge, HE1, 20MM, M56A1.

D7259302 — Security Range, Metal Parts.

LCA–SK–265 — Electronic Fragment Detection Screen.

LCA–SK–241 — Electronic Control, Gun Cut-Off.

F7548083 — Carton, Packing, Ammunition, for Fuze, PD, M505A1 Assembly, Details, and Packing.

F7548084 — Box, Packing, Ammunition, for Fuze, PD, M505A1 Assembly, Details, and Packing.

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PUBLICATIONS

ORDNANCE CORPS

ORD–M608–14 — Ammunition Ballistic Acceptance Test Methods, Part VIII.

ORD–M608–11 — Sampling Tables and Procedures for Continuous Sampling by Attributes.

GL 7259306 — Index to List of Inspection Gages and Parts Gaged On: Fuze, PD, M505A1, Assembly.

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein:

Interstate Commerce Commission

49 CFR 71–78—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

The ICC regulations are now a part of the Code of Federal Regulations (1949 Edition—Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950)."

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with their applicable drawings and specifications.

3.2 Fuze assembly. The fuze assembly, its metal parts and loaded components shall comply with the requirements of the applicable specifications and Drawing D7258818 and the drawings referenced thereon. (See 6.2).

3.3 Functioning. The fuze shall cause the projectile to function with high order detonation when tested as specified in 4.4.1.

3.4 Safety. There shall be no premature burst or fuze breakup, or evidence thereof, when the fuzes are tested as specified in 4.4.2.

3.5 Security. There shall be no separation of fuze metal parts in flight or in the gun when tested as specified in 4.4.2.

3.6 Stability. The fuze shall withstand rough handling without breakage or displacement of its metal components or without initiation of its explosive components when tested as specified in 4.4.3.

3.7 Workmanship. All parts shall be free from chips, dirt, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The provisions of Specification MIL–G–2550 apply to this specification. Reference will be made to MIL–STD–109 for definition of inspection terms used herein.

4.3 Sampling.

4.3.1 *Lot formation.* A lot shall consist of fuzes loaded by one manufacturer in accordance with the same drawing and drawing revision and same specification and specification revision, with the following limitations:

(a) Metal parts and metal parts assemblies of one interfix lot number from one manufacturer.

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(b) Detonators of one interfix lot number from one manufacturer.

(c) Boosters of one interfix lot number from one manufacturer.

4.3.2 *Examination.*

4.3.2.1 Each unit of the lot shall be examined for the critical defects listed in 4.3.4. Any unit found to be defective shall be removed from the lot.

4.3.2.2 Lot sampling and acceptance criteria will be in accordance with MIL–STD–105 or ORD–M608–11, as appropriate. Judgment for lot acceptance will be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.6):

4.3.2.2.1 *Individual defect characteristic basis.*

Major Defects	0.25%
Minor Defects	0.40%

4.3.2.2.2 *Defect characteristic class basis.*

Major Defects	1.50%
Minor Defects	2.50%

Individual defect characteristic AQL's associated with the class basis, will be

Major Defects	0.40%
Minor Defects	0.65%

4.3.2.3 Should components and subassemblies be furnished by the Government, the contractor shall ascertain by reference to data cards, description sheets or other inspection documents, prior to assembly of the fuzes, that such components and subassemblies have been inspected, tested and accepted in accordance with their respective drawings and specifications. In instances where the contractor also manufactures components or subassemblies for the fuze assembly, acceptance inspection and tests will be performed as prescribed in the pertinent detail specifications for the components or subassemblies.

4.3.3 *Test inspection.* The tests indicated below will be conducted in accordance with the methods specified in 4.4. Acceptance criteria will be as specified below.

4.3.3.1 *Functioning.* The sample fuze assemblies selected from each lot will be assembled into Cartridges, 20-mm, HE1, M56A1. A double sampling plan using the method of MIL–STD–105 will be applied. Sample sizes will be 50 and 100 fuze assemblies, respectively, with acceptance criteria of 1 and 3, respectively. Any assembly failing to function high order will be considered a defect with penalty assigned according to the acceptance criteria (see 6.4).

4.3.3.2 *Safety and security.* The sample fuze assemblies selected from each lot will be assembled into cartridges, 20mm, HE1, M56A1. A single sampling plan using the method of MIL–STD–105 will be applied. Sample sizes will be 50 fuze assemblies to be tested in each gun for which cartridge is standard. The acceptance number in this test is 0 and any premature burst or metal parts separation of the fuze components occurring either in the gun or in flight will be cause for rejection (see 6.3).

4.3.3.3 *Stability.* A single sampling plan utilizing the method of MIL–STD–105 will be applied to determine acceptability for fuze stability. Ten (10) fuze assemblies will be subjected to a jolt test and 10 fuze assemblies will be subjected to a jumble test. The acceptance number for each test is 0, and if there occurs any separation, deformation, displacement or breakage of the fuze components, or if any fuze assembly explodes during the test, the lot will be finally rejected. Where only external deformation of the external fuze surfaces occurs during the jumble test, it will be disregarded. Fuze assemblies subjected to these tests will be destroyed after completion of the tests.

4.3.4 *Classification of defects.*

4.3.4.1 *Rotor and detonator assembly (see Dwg. C7258829 covering a detail of Dwg. D7258818).*

Categories and defects	*Method of inspection*
Critical	
1. Detonator disc ruptured	Visual
2. Detonator inverted	Visual

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Categories and defects	Method of inspection
Major	
101. Detonator missing	Visual
102. Spherical surface of rotor distorted	Visual-Gage (see 4.3.5.1)
103. Detonator loose	Testing device
104. Sealant on sensitive end of detonator or on exterior surface or rotor	Visual
105. Crimp not full 360°	Visual
106. Crimp or detonator above reference surface	Gage
Minor	
None assigned.	

4.3.4.2 *Booster holder assembly. (See Dwg. C7258828 covering a detail of D7258818).*

Categories and defects	Method of inspection
Critical	
None assigned.	
Major	
101. Booster missing	Visual
102. Crimp not full 360°	Visual
103. Distortion of rotor seat	Visual-Gage (see 4.3.5.2)
104. Booster inverted	Visual
105. Foreign matter	Visual
106. Booster disc or cup ruptured	Visual
Minor	
201. Overall length	Gage
202. Booster loose	Approved test

4.3.4.3 *Assembly, prior to assembly of cover. (See Dwg. D7258818).*

Categories and defects	Method of inspection
Critical	
1. Rotor improperly positioned	Visual (see 4.3.5.3)
2. Spring missing or distorted	Visual (see 4.3.5.3)
3. Firing pin flange missing	Visual
Major	
101. Firing pin missing	Visual
102. Spring retainer missing	Visual
103. Firing pin damaged, or improperly formed	Visual
Minor	
None assigned.	

4.3.4.4 *Assembly. (See Dwg. D7258818).*

Categories and defects	Method of inspection
Critical	
None assigned.	
Major	
101. Cracked, split or deformed cover	Visual
102. Cover loose	Testing device (see 4.3.5.4)
103. Booster holder loose	Testing device (see 4.3.5.5)
104. Length from shoulder to end of booster holder	Gage
Minor	
201. Length from shoulder to nose.	Gage
202. Sealing missing or improper...	Visual (see 4.3.5.6)
203. Cover crimp not full 360°	Visual
204. Booster holder stake marks missing	Visual
205. External thread damaged	Visual-Gage
206. Plating rusted or discolored...	Visual

4.3.4.5 *Packed carton (see Dwg. F7548083).*

Categories and defects	Method of inspection
Critical	
None assigned.	
Major	
101. Incorrectly closed and secured carton	Visual (see 4.3.5.7)
102. Marking incorrect, incomplete or unidentifiable	Visual
103. Carton damaged	Visual (see 4.3.5.8)
104. Contents short	Count
Minor	
201. Marking touched up or repaired with non-permissible material (crayon, chalk, etc.)	Visual

4.3.4.6 *Sealed packing box (see Dwg. F7548084).*

Categories and defects	Method of inspection
Critical	
None assigned.	
Major	
101. Box damaged	Visual (see 4.3.5.9)
102. Marking incomplete, incorrect or unidentifiable	Visual
103. Contents short	Count
Minor	
201. Marking touched up or repaired with non-permissible material (crayon, chalk, etc.)	Visual

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4.3.5 *Defect standards.*

4.3.5.1 Any rotor containing flats, burrs, nicks, or distortion of the spherical surface as determined by visual examination or which fails to pass the specified diameter gage will be classed defective.

4.3.5.2 Any booster holder in which the spherical surface of the rotor seat is distorted as determined by visual examination, or fails to pass the specified gages, "Length from forward edge of flange to top of .37700 ball" and "Concentricity of .37700 ball with pitch diameter of thread," will be classed defective.

4.3.5.3 Any assembly in which the rotor safety spring does not seat firmly against the flat surface of the rotor or in which the end of the spring is not properly positioned against the firing pin will be classed defective.

4.3.5.4 Any movement between the fuze body and cover will be classed as a defect.

4.3.5.5 Any movement between the fuze body and booster holder will be classed as a defect.

4.3.5.6 Coating missing from any portion of the joined surfaces, as determined by visual methods, will be classed defective.

4.3.5.7 Any carton on which the sealing tape is inadequate or which does not adhere to the carton will be classed defective.

4.3.5.8 "Carton damaged" is defined as any tear, perforation, water damage or other defect which would expose contents or permit or indicate damage to the contents.

4.3.5.9 "Box damage" is described as any split, knotholes, broken or misplaced hardware or missing cleat.

4.3.6 *Inspection equipment.*

4.3.6.1 Government verification inspection will be accomplished with the use of gages and equipment specified in the Index to Lists of Inspection Gages and Parts Gaged On: Fuze, PD, M505A1 Assembly, GL7259306.

4.4 *Test methods.*

4.4.1 *Functioning.* Functioning tests will be conducted in accordance with Army Ordnance Publication, ORD-M608-14—Ammunition Ballistic Acceptance Test Methods, Part VIII, Chapter 1, Section 7. Impact of the projectile will be against an aluminum target (2024T4, T3 or T36) of .040 nominal thickness placed 200 yards ±25 yards from the gun muzzle normal to the line of fire. Observation will be made from a position at the side of and slightly down range from the target to assure an accurate determination of the point of functioning and character of detonation (see 6.5).

4.4.2 *Safety and security.* The test procedure will be as prescribed in Army Ordnance Publication, ORD-M608-14—Ammunition Ballistic Acceptance Test Methods, Part VIII, Chapter 1, Section 7. The firing will be accomplished on a range set up as shown on Drawing D7259302 using Electronic Fragment Detection Screen, Dwg. LCA-SK-265, and Electronic Control Gun Cut-Off, Drawing LCA-SK-241.

4.4.3 *Stability tests.* These tests (Jolt and Jumble) will be conducted in accordance with the methods and utilizing the equipment specified in Military Standards 350 and 351, respectively.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing.

5.1.1 *Level C.* Two hundred assemblies shall be packaged in a corrugated fiberboard carton in two layers in a 10 x 10 arrangement per layer, in accordance with the requirements of Dwg. F7548083.

5.2 Packing and marking.

5.2.1 *Level C.* One thousand two hundred (1,200) assemblies (in six cartons) shall be packed in a wood packing box in accordance with the requirements of Dwg. F7548084. Marking shall be as specified on the drawing, and as prescribed by ICC publication 49 CFR 71-78, and to the extent applicable as prescribed by MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Prior to assembly of detonators and boosters into fuze lots, it will be ascertained that the elapsed time from the original acceptance of the detonators and boosters has not exceeded 180 days. If this limit has been exceeded, the detonators and boosters will be tested for and must pass the functioning and output acceptance criteria of their respective detail specifications. Detonator or booster lots failing to pass these tests will be referred to the procuring activity for disposition.

6.3 A premature burst is defined as functioning of the fuze in the bore of the gun, or in flight before impact.

6.4 High order detonation is defined as follows: Detonation which produces a brilliant flash 4 to 6 feet in diameter, accompanied by a violent sharp concussion producing report.

6.5 Photographic evidence of the character of detonation should be made for each fuze tested in conjunction with visual observation. Photographs to be used when quality of detonation is suspect.

6.6 AQL's. The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is authorized to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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